

Part A:-

1)  $255.375 \rightarrow$  octal, binary / hexa  
 $255.375$

Binary :-

|                       |                                       |
|-----------------------|---------------------------------------|
| $255/2 \rightarrow 1$ | $0.375 \times 2 = 0.75 \rightarrow 0$ |
| $127/2 \rightarrow 1$ | $0.75 \times 2 = 1.5 \rightarrow 1$   |
| $63/2 \rightarrow 1$  | $0.5 \times 2 = 1 \rightarrow 1$      |
| $31/2 \rightarrow 1$  |                                       |
| $15/2 \rightarrow 1$  |                                       |
| $7/2 \rightarrow 1$   |                                       |
| $3/2 \rightarrow 1$   |                                       |
| $1/2 \rightarrow 1$   |                                       |

So,  $(255.375)_{10} = (11111111.011)_2$

Octal :-

|                       |                      |
|-----------------------|----------------------|
| $255/8 \rightarrow 7$ | $0.375 \times 8 = 3$ |
| $31/8 \rightarrow 7$  |                      |
| $3/8 \rightarrow 3$   |                      |

$(255.375)_{10} = (377.3)_8$

$\rightarrow (255.375)_{10} = \left( \begin{array}{l} 255 \\ 16 \end{array} \rightarrow 15 \rightarrow F \right) \left( \begin{array}{l} 0.375 \times 16 \rightarrow 6 \end{array} \right)$   
 $\left( \begin{array}{l} 15 \\ 16 \end{array} \rightarrow 15 \rightarrow F \right)$   
 $(255.375)_{10} = (FF.6)_{16}$

2) Decimal value :-

$(110101.10)_2$

$1 \times 2^{-1} + 0 \times 2^{-2} + 1 \times 2^{-3}$

$0.5 + 0.125$

$= 0.625$

↓  
decimal value

3) Even powers give +1 as remainder  
 1, ' ' ' +2 as remainder

100111 has equal no. of odd powers  
 & even powers.

So, then we can say that it is divisible  
 by 3.

$$\begin{array}{rcl} 23 & \div & 1 \\ \hline 12 & \div & 0 \\ \hline 8 & \div & 0 \\ \hline 2 & \div & 1 \\ \hline 0 & & \end{array}$$

4) -23 (to write -23, write 23 & flip to digits)

$$23 = (00110011)_2$$

$$\boxed{-23 = (11101100)_2}$$

5)  $F = (A+B)(A'+C)(B+C')$

It is a series of 'or' & 'and' logic gates.

- 1st  $\Rightarrow A$  or  $B$  should be true  
 2nd  $\Rightarrow A'$  or  $C$  should be true  
 3rd  $\Rightarrow B$  or  $C'$  should be true
- } all need to satisfy

6)  $F(A,B,C) = \sum m(1,3,5,7)$

$m \Rightarrow$  minterms state that the output is 1 for a specific combination of variables & 0 for rest.

So, at  $m_1, m_3, m_5, m_7 \rightarrow 1$   
 rest all  $\rightarrow 0$